YouTube-DNN-RecSys — 系统架构 · 召回数据流

Python / Spark data preparation → TFRecords → dnn.py → user & item vectors → prediction

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

项目介绍

YouTube-DNN-RecSys 使用 TensorFlow 与 Spark 实现 YouTube 推荐系统中的候选召回阶段。

系统将用户行为历史转换为 TFRecords，训练用户与物品向量，并为在线服务生成 Top-K 召回候选。

DATA PROCESSING · 离线样本构建

Python Pipeline
data2tfrecords.py
read_tfrecords.py

Spark Pipeline
MakeDataOne.scala
MakeDataTwo.scala

FeatureBuilder
history + continuous features

STORAGE · 训练与模型产物

TFRecords
train / evaluation

Checkpoints
ckpt / eval / modelpath

MODEL TRAINING · src/python/models/dnn.py

Input
visit_items_index × 5
continuous_features × 16

Item Embedding
500 × 16 · average pooling

User Tower
Dense 64 → 32 → vector 16

Softmax Output
500 classes · Top-K

evaluation feedback

PREDICTION · serving paths

Python SavedModel
load_dnn_model.py

User Vector
PredictUserVector.scala

Item Table
ItemEmbeddingPredictor.scala

RECOMMENDATION OUTPUT

Candidate Recall
user vector × item embedding

Top-K Items
next-item recommendation

主链：行为历史 → TFRecords → embedding mean-pool → user vector → item table matching → Top-K。